

20.2 Electromagnetic Induction

Question Paper

Course	CIE A Level Physics
Section	20. Magnetic Fields
Topic	20.2 Electromagnetic Induction
Difficulty	Easy

Time allowed: 50

Score: /38

Percentage: /100

Question 1a

State Faraday's Law

(i)

in words

[2]

(ii)

as an equation, defining all the quantities.

[2]

[4 marks]**Question 1b**

Lenz's law is sometimes combined with Faraday's law in order to explain electromagnetic effects.

Use the words below to complete the description of Lenz's law

attracts opposes density direction linkage magnitude
--

The _____ of an induced emf is always set up in a way such that it _____ the change in magnetic flux _____ that causes it.

[3 marks]

Question 1c

Rewrite the equation you wrote in **(a)(ii)** to include Lenz's law.

[1 mark]

Question 1d

A small coil of wire connected to a sensitive voltmeter is placed near one end of a solenoid (in which an electrical current is flowing) as shown in Fig. 1.1.

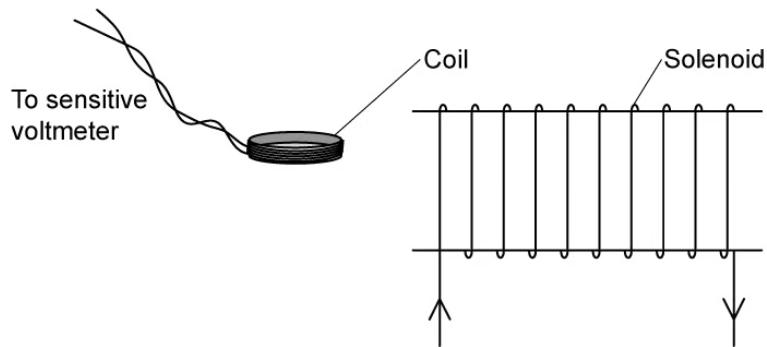


Fig. 1.1

State **two** different ways in which an e.m.f. may be induced in the coil.

[2 marks]

Question 2a

(i)

Define magnetic flux.

[2]

(ii)

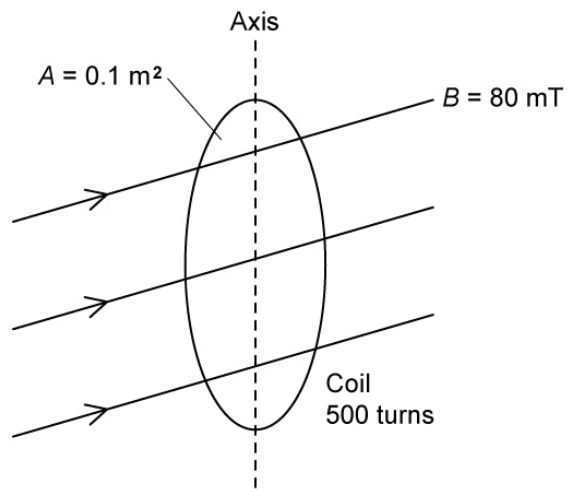
Write the equation for magnetic flux when the magnetic flux density is **not** perpendicular to the area.

[1]

[3 marks]

Question 2b

A circular coil with an area of 0.1 m^2 and 500 turns is placed with its plane perpendicular to a horizontal magnetic field of uniform flux density 80 mT as shown in Fig. 1.1.

**Fig. 1.1**

Calculate the magnetic flux passing through the coil. Give a unit with your answer.

[3 marks]

Question 2c

The coil experiences a change in magnetic flux linkage when it is rotated through 90° about a vertical axis (indicated in Fig. 1.1).

(i)

Define magnetic flux linkage

[1]

(ii)

Calculate the change in magnetic flux linkage produced by the rotation of the coil.

[2]

[3 marks]

Question 2d

The coil rotates through 90° in a time of 300 ms.

Calculate the average magnitude of the induced e.m.f. in the coil during the rotation.

[3 marks]

Question 3a

Table 1.1 shows the standard international (SI) units of the main quantities involved in electromagnetic induction.

Table 1.1

Quantity	Symbol	SI unit
magnetic flux	ϕ	
magnetic flux linkage		Wb turns
electromotive force		
	B	

Complete Table 1.1 by filling in the missing quantities, symbols and SI units.

[4 marks]

Question 3b

A coil is connected to a voltmeter, which is centred at zero as shown in Fig. 1.1a.

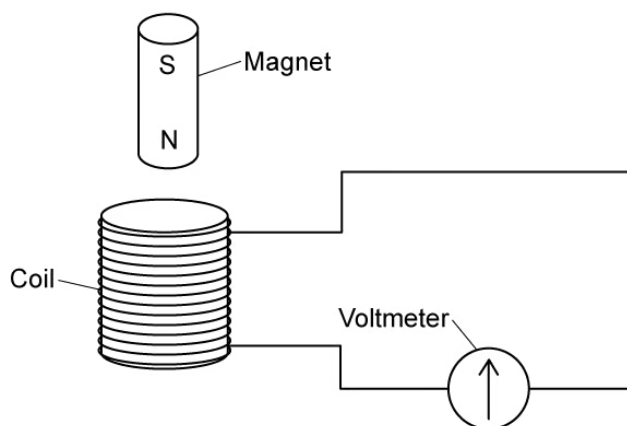


Fig. 1.1a

When the magnet is lowered into the coil, the needle on the voltmeter deflects to the right as shown in Fig. 1.1b.



Fig. 1.1b

On Fig. 1.1c, sketch the expected observations of the voltmeter needle when

(i)
the magnet is held at rest in the coil

(ii)
the magnet is removed from the coil faster than it entered



Fig. 1.1c

[1]

[3]

[4 marks]

Question 3c

The magnet is moved towards and away from the coil at a steady rate.

Fig. 1.2 shows the variation of time with the reading of potential difference on the voltmeter.

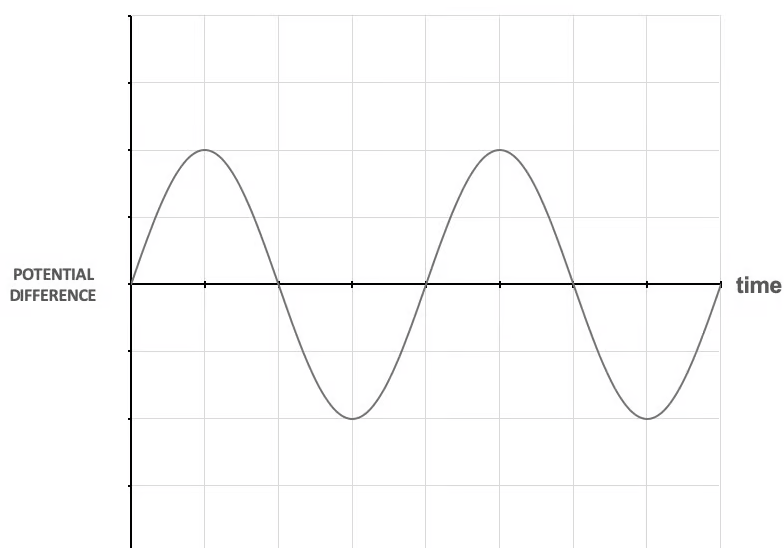


Fig. 1.2

Use Fig. 1.2 and Faraday's law to explain why

- (i)
there is a reading on the voltmeter, [1]
- (ii)
this reading varies in magnitude, [2]
- (iii)
the reading has both positive and negative values. [2]

[5 marks]

Question 3d

As the magnet is moved away from the coil, the voltmeter shows a value of 1.5 mV for 2.0 s.

Calculate the change in magnetic flux linkage as the magnet is moved away from the coil.

[3 marks]

Model Answers: Easy

1a

(a)

(i) Faraday's law of electromagnetic induction states:

- The (magnitude of the) induced e.m.f. is (directly) proportional to the rate; [1 mark]
- of change of magnetic flux linkage; [1 mark]

(ii) Faraday's law is defined by the equation

- $\varepsilon = \frac{\Delta(N\Phi)}{\Delta t}$ [1 mark]

- Where ε = induced e.m.f., $\Delta(N\Phi)$ = change in magnetic flux linkage and Δt = time interval of change; [1 mark]

[Total: 4 marks]

It's crucial that you remember an emf is only induced if there is a **change** in magnetic flux linkage.

How quickly this quantity changes (i.e., the **rate** of change of magnetic flux linkage) determines the magnitude of the induced emf.

1b

(b) The correct sentence is:

- The **direction** of an induced emf; [1 mark]
- is always set up in a way such that it **opposes**; [1 mark]
- the change in magnetic flux **linkage** that causes it; [1 mark]

[Total: 3 marks]

Stating Lenz's law in words comes up quite often in examinations, make sure you can

describe it in terms of the **direction** of the induced emf

The induced emf is always set up in such a way that the effect is to try and **stop**, or **oppose**, whatever is inducing it. This is a consequence of **conservation of energy** (i.e., electromagnetic energy cannot be created for free!).

1c

(c) The equation including Lenz's law is:

- $\varepsilon = - \frac{\Delta(N\Phi)}{\Delta t}$
- Minus sign included; [1 mark]

[Total: 1 mark]

Lenz's law says the **direction** of the induced emf is set up to oppose the change causing it, which is encapsulated by the negative sign in the equation.

This is similar to vectors, for example, if vector **a** and vector **b** are related by **a = - b**, this means they point in opposite directions. In this sense, the vectors '**oppose**' each other, just like the induced emf ε opposes the changing flux linkage.

1d

(d) An e.m.f. may be induced in the coil by:

Any **two** from:

- Moving the coil towards / away from the solenoid; [1 mark]
- Rotating the coil; [1 mark]
- Varying the current in the solenoid; [1 mark]
- Placing an iron core into the solenoid; [1 mark]

[Total: 2 marks]

2a

(a)

(i) Magnetic flux is defined as:

- The product of (magnetic) flux density B and area A ; [1 mark]
- Where the area is perpendicular to the (magnetic) field; [1 mark]

OR

- Magnetic flux: $\Phi = BA$; [1 mark]
- Where B = magnetic flux density and A = cross-sectional area (through the coil); [1 mark]

(ii) The equation for magnetic flux when B and A are not perpendicular is:

- $\Phi = BA \cos \theta$; [1 mark]

[Total: 3 marks]

2b

(b) Calculate the magnetic flux passing through the coil:

List the known quantities:

- Cross-sectional area, $A = 0.1 \text{ m}^2$
- Magnetic flux density, $B = 80 \text{ mT} = 80 \times 10^{-3} \text{ T}$

Calculate the magnetic flux and state the unit:

- Magnetic flux: $\Phi = BA$
- $\Phi = (80 \times 10^{-3}) \times 0.1$ [1 mark]
- $\Phi = 8 \times 10^{-3}$ [1 mark] Wb [1 mark]

[Total: 3 marks]

2c

(c)

(i) Magnetic flux linkage is defined as:

- The product of the magnetic flux and the number of turns on a coil ($N\Phi$) **OR** the

product of magnetic flux density, area, and the number of turns on a coil (BAN); [1 mark]

(ii) Calculate the change in magnetic flux linkage:

List the known quantities:

- Magnetic flux, $\Phi = 8 \times 10^{-3}$ Wb (from **(b)**)
- Number of turns, $N = 500$

Calculate the change in magnetic flux linkage:

- Flux linkage: $N\Phi = BAN$
- Change in flux linkage: $\Delta(N\Phi) = 500 \times (8 \times 10^{-3}) - 0$ [1 mark]
- $\Delta(N\Phi) = 4$ Wb (turns) [1 mark]

[Total: 3 marks]

2d

(d) Calculate the average magnitude of the induced e.m.f. in the coil during the rotation:

List the known quantities:

- Change in flux linkage, $\Delta(N\Phi) = 4$ Wb
- Time interval, $\Delta t = 300$ ms = 300×10^{-3} s

Use Faraday's law to calculate the induced e.m.f.

- Faraday's law: $\varepsilon = \frac{\Delta(N\Phi)}{\Delta t}$ [1 mark]
- Induced e.m.f. $\varepsilon = \frac{4}{300 \times 10^{-3}}$ [1 mark]
- Induced e.m.f. $\varepsilon = 13.3 \approx 13$ V [1 mark]

[Total: 3 marks]

2d

(d) Calculate the average magnitude of the induced e.m.f. in the coil during the rotation:

List the known quantities:

- Change in flux linkage, $\Delta(N\Phi) = 4 \text{ Wb}$
- Time interval, $\Delta t = 300 \text{ ms} = 300 \times 10^{-3} \text{ s}$

Use Faraday's law to calculate the induced e.m.f.

- Faraday's law: $\varepsilon = \frac{\Delta(N\Phi)}{\Delta t}$ [1 mark]
- Induced e.m.f. $\varepsilon = \frac{4}{300 \times 10^{-3}}$ [1 mark]
- Induced e.m.f. $\varepsilon = 13.3 \approx 13 \text{ V}$ [1 mark]

[Total: 3 marks]

3a

(a) The correct table is as follows:

Quantity	Symbol	SI unit	
magnetic flux	ϕ	Wb / Weber	[1 mark]
magnetic flux linkage	$N\Phi$	Wb turns	[1 mark]
electromotive force	ε	V / Volt	[1 mark]
magnetic flux density	B	T / Tesla	[1 mark]

[Total: 4 marks]

3b

(b) The correct diagrams are as follows:



(i) If the magnet is held at rest in the coil:

- No deflection shown (arrow still pointing upwards); [1 mark]

(ii) If the magnet is moved back out of the coil more quickly than it entered:

- Deflection shown; [1 mark]
- To the left; [1 mark]
- Greater than the deflection shown in Fig. 1.1b; [1 mark]

[Total: 4 marks]

The deflection on the voltmeter is a **measure** of the magnitude of the **induced e.m.f.**

When the magnet is held **at rest** inside a coil, there is no **change in** magnetic flux linkage, therefore there is no induced e.m.f. or current (and no deflection on the voltmeter).

If the magnet reverses from the coil, there **is** a change in magnetic flux linkage in the **opposite** direction to that initially observed in **Fig. 1.1b** – therefore the deflection on the voltmeter is in the opposite direction as the current is in the opposite direction.

Notice the question says the magnet is removed **faster** than it entered, therefore, the **rate of change** of magnetic flux linkage is greater which means a larger induced e.m.f. and current (and larger deflection), as required by Faraday's law.

3c

(c) Using the graph and Faraday's law...

- Faraday's law: induced e.m.f. is proportional to the rate of change of magnetic flux linkage

(i) There is a reading on the voltmeter because:

- This causes a change in flux linkage; [1 mark]

(ii) The reading varies in magnitude because:

- The speed of the magnet varies; [1 mark]
- This causes a variation in the rate of change of flux; [1 mark]

(iii) The reading has both positive and negative values because

- The magnet changes its direction; [1 mark]
- This causes the induced e.m.f. / current to change direction; [1 mark]

[Total: 5 marks]

3d

(d) Calculate the change in magnetic flux linkage:

List the known quantities:

- Induced e.m.f., $\varepsilon = 1.5 \text{ mV} = 1.5 \times 10^{-3} \text{ V}$
- Time interval, $t = 2.0 \text{ s}$

Use Faraday's law and rearrange:

- $\varepsilon = \frac{\Delta(N\Phi)}{\Delta t} \Rightarrow \Delta(N\Phi) = \varepsilon \Delta t$ [1 mark]
- $\Delta(N\Phi) = (1.5 \times 10^{-3}) \times 2.0$ [1 mark]
- $\Delta(N\Phi) = 3.0 \times 10^{-3} \text{ Wb turns}$ [1 mark]

[Total: 3 marks]

Remember that magnetic flux linkage has units Wb (Weber) turns, this is a commonly forgotten unit!